

Newton's Laws of Motion — Complete Notes

Grade 11

Physical Sciences

CAPS

What you'll learn

- State and explain Newton's three laws of motion in your own words.
- Tell the difference between mass and weight, and between contact and non-contact forces.
- Draw free-body diagrams and choose a positive direction before calculating.
- Use $F_{\text{net}} = m a$ to find acceleration, net force, mass or a single force.
- Work with the normal force and with friction (static and kinetic) on flat surfaces.

What is a force?

A force is a push or a pull that one object exerts on another. It is a vector quantity, so it has both magnitude and direction. The SI unit of force is the newton (N), where $1 \text{ N} = 1 \text{ kg} \cdot \text{m} \cdot \text{s}^{-2}$.

Forces can be grouped as:

- Contact forces — normal force, friction, tension, an applied push or pull.
- Non-contact (field) forces — weight (gravity), magnetic force, electrostatic force.

When several forces act on one object, we add them as vectors to get the net force (also called the resultant force), F_{net} .

Newton's first law (the law of inertia)

An object continues in its state of rest, or in uniform motion in a straight line, unless acted on by a non-zero net force.

- Inertia is the resistance of an object to any change in its state of motion.
- Mass is the measure of an object's inertia: a larger mass means larger inertia and a smaller acceleration for the same force.
- "At rest" and "moving at constant velocity" are physically the same state — in both cases the acceleration is zero and $F_{\text{net}} = 0$.
- This is why a passenger without a seatbelt keeps moving forward when a car brakes suddenly.

Newton's second law

When a non-zero net force acts on an object, the object accelerates in the direction of the net force. The acceleration is directly proportional to the net force and inversely proportional to the mass of the object.

$$F_{\text{net}} = m a$$

Symbol	Meaning	SI unit
F_{net}	net (resultant) force	N
m	mass	kg
a	acceleration	$\text{m} \cdot \text{s}^{-2}$

Rearranged: $a = F_{\text{net}} / m$ and $m = F_{\text{net}} / a$. Always choose a positive direction first, then substitute the signs carefully.

Newton's third law

If object A exerts a force on object B, then object B simultaneously exerts a force on object A that is equal in magnitude and opposite in direction.

$F_{A \text{ on } B} = - F_{B \text{ on } A}$

- The two forces form an action–reaction pair.
- They act on different objects, so they never cancel each other out for a single object.
- Example: a swimmer pushes water backwards; the water pushes the swimmer forwards.

Mass and weight

Mass and weight are not the same.

Quantity	Symbol	Unit	Type	Formula
Mass	m	kilogram (kg)	scalar (measure of inertia)	—
Weight	w or F_g	newton (N)	vector, always downwards	$w = m g$

On Earth $g \approx 9,8 \text{ m} \cdot \text{s}^{-2}$. A 70 kg learner has a weight of $w = 70 \times 9,8 = 686 \text{ N}$. Mass stays the same everywhere; weight changes if g changes (for example on the Moon).

Free-body diagrams

A free-body diagram shows only the forces acting on one chosen object, drawn as arrows from a dot.

N (normal force, up)

↑

|

f ←-- [m] --→ F (applied force)

|

↓

w = m g (weight, down)

Rules:

- Draw all the forces acting ON the object.
- Do not draw the forces the object exerts on other things.
- Label every arrow with its symbol.
- Choose a positive direction (usually right = + or up = +) before doing the maths.

Normal force and friction

The normal force N is the perpendicular force a surface exerts on an object resting on it. On a horizontal surface with no other vertical forces, $N = m g$.

Friction opposes relative motion (or attempted motion) between two surfaces in contact.

- Static friction f_s prevents motion from starting. Its maximum value is $f_s(\text{max}) = \mu_s N$.
- Kinetic (dynamic) friction f_k acts while the object slides: $f_k = \mu_k N$.
- μ_s is usually greater than μ_k , which is why it takes more force to start an object moving than to keep it moving.

Worked example 1 — horizontal motion with friction

A 6,0 kg crate is pulled along a rough horizontal floor by a horizontal applied force of 30 N. The coefficient of kinetic friction between the crate and the floor is 0,25. Calculate the acceleration of the crate.

Given: $m = 6,0 \text{ kg}$; $F = 30 \text{ N}$; $\mu_k = 0,25$; $g = 9,8 \text{ m}\cdot\text{s}^{-2}$. To find: a .

Step 1 — Vertical: the crate does not move up or down, so the vertical forces balance.

$$N = m g = 6,0 \times 9,8 = 58,8 \text{ N}$$

Step 2 — Kinetic friction:

$$f_k = \mu_k N = 0,25 \times 58,8 = 14,7 \text{ N}$$

Step 3 — Horizontal net force (take right as positive):

$$F_{\text{net}} = F - f_k = 30 - 14,7 = 15,3 \text{ N}$$

Step 4 — Newton's second law:

$$a = F_{\text{net}} / m = 15,3 / 6,0 = 2,55 \text{ m}\cdot\text{s}^{-2}$$

Answer: the crate accelerates at $2,55 \text{ m}\cdot\text{s}^{-2}$ in the direction of the pull.

Worked example 2 — lift (vertical motion)

A 60 kg passenger stands on a bathroom scale inside a lift. The lift accelerates upwards at $2,0 \text{ m}\cdot\text{s}^{-2}$. Calculate the reading on the scale.

The scale reading equals the normal force N of the scale on the passenger.

Given: $m = 60 \text{ kg}$; $a = 2,0 \text{ m}\cdot\text{s}^{-2}$ (upwards); $g = 9,8 \text{ m}\cdot\text{s}^{-2}$. Take upwards as positive. The forces are N up and the weight $w = m g$ down.

$$F_{\text{net}} = N - m g = m a$$

$$N = m g + m a = m (g + a)$$

$$N = 60 \times (9,8 + 2,0) = 60 \times 11,8 = 708 \text{ N}$$

Worked example 3 — connected objects

A 4,0 kg block on a frictionless horizontal table is connected by a light, inextensible string over a frictionless pulley to a hanging 2,0 kg mass. Calculate the acceleration of the system and the tension in the string.

Given: $m_1 = 4,0 \text{ kg}$ (on the table); $m_2 = 2,0 \text{ kg}$ (hanging); $g = 9,8 \text{ m}\cdot\text{s}^{-2}$.

Step 1 — For the hanging mass m_2 (taking downwards as positive for it):

$$m_2 g - T = m_2 a \dots (1)$$

Step 2 — For m_1 on the table (horizontal, no friction):

$$T = m_1 a \dots (2)$$

Step 3 — Add (1) and (2):

$$m_2 g = (m_1 + m_2) a$$

$$a = m_2 g / (m_1 + m_2) = (2,0 \times 9,8) / (4,0 + 2,0) = 19,6 / 6,0 = 3,27 \text{ m}\cdot\text{s}^{-2}$$

Step 4 — Tension:

$$T = m_1 a = 4,0 \times 3,27 = 13,1 \text{ N}$$

Check with (1): $m_2 g - T = 19,6 - 13,1 = 6,5 \text{ N}$, and $m_2 a = 2,0 \times 3,27 = 6,5 \text{ N} \checkmark$

Common mistakes

- Confusing mass (kg) and weight (N). Weight is a force: $w = m g$, always downwards.
- Forgetting that Newton's third-law pairs act on different objects and therefore never cancel on a single free-body diagram.
- Using $g = 10 \text{ m}\cdot\text{s}^{-2}$ when the question expects $9,8 \text{ m}\cdot\text{s}^{-2}$ (or the other way around). Use what the paper gives; otherwise use $9,8 \text{ m}\cdot\text{s}^{-2}$.
- Ignoring the direction of forces, so the signs in $F_{\text{net}} = m a$ come out wrong. Choose a positive direction and stick to it.
- Writing " $F = m a$ " when you mean the net force. An applied force on its own is not F_{net} .
- Assuming an object at rest has no forces on it. It has balanced forces, so $F_{\text{net}} = 0$.
- Mixing units, for example grams with kilograms or cm with m. Convert first.

Quick summary / cheat sheet

- First law: $F_{\text{net}} = 0 \rightarrow$ the object stays at rest or moves at constant velocity (zero acceleration).
- Second law: $F_{\text{net}} = m a \rightarrow$ the acceleration is in the direction of the net force.
- Third law: forces come in equal-and-opposite pairs acting on different objects.
- Weight: $w = m g$, with $g = 9,8 \text{ m}\cdot\text{s}^{-2}$ near the Earth's surface.
- Normal force on a flat surface with no other vertical forces: $N = m g$.
- Friction: $f_k = \mu_k N$ (sliding); maximum static friction $f_s(\text{max}) = \mu_s N$.
- Always: choose a positive direction, draw a free-body diagram, write $F_{\text{net}} = m a$, then substitute with units.

Practice questions

1. State Newton's first law of motion. [2]
2. A 1 200 kg car accelerates from rest to a speed of $20 \text{ m}\cdot\text{s}^{-1}$ in 8,0 s on a level road. Ignoring friction, calculate the net force on the car. [3]
3. A 5,0 kg box rests on a horizontal floor. The coefficient of static friction is 0,40. Calculate the minimum horizontal force needed to start the box moving. [3]
4. A 0,50 kg ball and a 5,0 kg bag are pushed with the same net force of 10 N. Which one has the greater acceleration, and by how much? [3]
5. A 75 kg astronaut has a weight of 736 N on a planet. Calculate the gravitational acceleration g on that planet. [2]
6. A 3,0 kg block hangs from a string inside a lift that accelerates downwards at $1,5 \text{ m}\cdot\text{s}^{-2}$. Calculate the tension in the string. [3]

7. Explain, using Newton's third law, why a rocket can move in outer space where there is no air to push against. [3]

Answers

1. An object continues in its state of rest, or in uniform motion in a straight line, unless acted on by a non-zero net force.

1. $a = (v - u) / t = (20 - 0) / 8,0 = 2,5 \text{ m} \cdot \text{s}^{-2}$

$F_{\text{net}} = m a = 1\,200 \times 2,5 = 3\,000 \text{ N} = 3,0 \times 10^3 \text{ N}.$

1. $N = m g = 5,0 \times 9,8 = 49 \text{ N}$

$f_s(\text{max}) = \mu_s N = 0,40 \times 49 = 19,6 \text{ N} \approx 20 \text{ N}$ The minimum applied force $\approx 20 \text{ N}$ (it must just exceed the maximum static friction).

1. $a_{\text{ball}} = 10 / 0,50 = 20 \text{ m} \cdot \text{s}^{-2}$; $a_{\text{bag}} = 10 / 5,0 = 2,0 \text{ m} \cdot \text{s}^{-2}.$

The ball accelerates more, by $20 - 2,0 = 18 \text{ m} \cdot \text{s}^{-2}$ (ten times more, since its mass is ten times smaller).

1. $w = m g \rightarrow g = w / m = 736 / 75 = 9,81 \text{ m} \cdot \text{s}^{-2} \approx 9,8 \text{ m} \cdot \text{s}^{-2}.$

1. Take downwards as positive for the hanging block: $m g - T = m a$

$T = m (g - a) = 3,0 \times (9,8 - 1,5) = 3,0 \times 8,3 = 24,9 \text{ N} \approx 25 \text{ N}.$

1. The rocket pushes hot gases backwards; by Newton's third law the gases push the rocket forwards with an equal and opposite force. No air is needed — the action–reaction pair is between the rocket and its own exhaust gases.

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